

Linear Algebra

2 Vector spaces

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Central idea: adding and multiplying by scalars — always with the same rules — works for arrows, n -tuples, polynomials and functions.

Exercises: Sheet 2, Ex. 16–38 (original numbering; Ex. 15 was removed).

2.1 Vector space

What do they have in common?

- $(1, 2, 3) + (0, 1, 2) = (1, 3, 5),$
 $2(1, 2, 3) = (2, 4, 6)$
- $(1 + x) + (x - x^2) = 1 + 2x - x^2$
- $(\sin x) + (2 \cos x), \quad 3e^x$ (functions)
- $(2 + i) + (1 - 3i) = 3 - 2i,$
 $2(2 + i) = 4 + 2i$

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In every case we know how to

- **add** two objects and
- **multiply** an object by a number,

and the rules of calculation are *the same*.

A **vector space** is any set where this happens.
Its elements are called **vectors**, even if they
are polynomials or functions.

Definition

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$ (the **scalars**). A set U with an *addition* $\mathbf{u} + \mathbf{v} \in U$ and a *scalar multiplication* $\alpha \mathbf{u} \in U$ is a **vector space over $\mathbb{K}$** if, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in U$ and $\alpha, \beta \in \mathbb{K}$:

Addition

$$\text{A1 } (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

$$\text{A2 } \mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$\text{A3 } \text{there is } \mathbf{0} \in U \text{ with } \mathbf{u} + \mathbf{0} = \mathbf{u}$$

$$\text{A4 } \text{there is } -\mathbf{u} \in U \text{ with } \mathbf{u} + (-\mathbf{u}) = \mathbf{0}$$

Scalar multiplication

$$\text{M1 } \alpha(\mathbf{u} + \mathbf{v}) = \alpha \mathbf{u} + \alpha \mathbf{v}$$

$$\text{M2 } (\alpha + \beta)\mathbf{u} = \alpha \mathbf{u} + \beta \mathbf{u}$$

$$\text{M3 } \alpha(\beta \mathbf{u}) = (\alpha\beta)\mathbf{u}$$

$$\text{M4 } 1 \mathbf{u} = \mathbf{u}$$

$\mathbb{K} = \mathbb{R}$: *real* vector space; $\mathbb{K} = \mathbb{C}$: *complex*. It is also called a *linear space*.

Examples and non-examples

Examples (with the usual operations):

- $\mathbb{R}^n$ and $\mathbb{C}^n$ (sum and product coordinate by coordinate);
- P_n : real polynomials of degree $\leq n$;
- real functions with domain $\mathbb{R}$: $(f + g)(x) = f(x) + g(x)$, $(\alpha f)(x) = \alpha f(x)$;
- $\mathbb{C}$ as a *real* vector space (Ch. 1: $a + bi$ behaves like the pair (a, b)).

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Non-examples

- Polynomials of degree *exactly* 2: $(x^2 + x) + (-x^2) = x$ has degree 1 (the sum leaves the set).
- $\{(x, y) : y = x + 1\}$: does not contain $(0, 0)$.

Consequences of the axioms

For any $\mathbf{u} \in U$ and $\alpha \in \mathbb{K}$:

$$0\mathbf{u} = \mathbf{0}, \quad \alpha\mathbf{0} = \mathbf{0}, \quad (-1)\mathbf{u} = -\mathbf{u},$$

$$\alpha\mathbf{u} = \mathbf{0} \iff \alpha = 0 \text{ or } \mathbf{u} = \mathbf{0}.$$

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To show that it is *not* a vector space

One axiom that fails, with **one** concrete example, is enough. To show that it *is*, all of them must be checked.

- **Ex. 16** $\mathbb{K}^n$ is a vector space (check the axioms).
- **Ex. 17** P_2 is a vector space.
- **Ex. 18** ‘strange’ operations on $\mathbb{R}^3$: find the axiom that fails.

2.2 Linear combination and subspaces

Definition

A **linear combination** of $\mathbf{u}_1, \dots, \mathbf{u}_n \in U$ is any vector

$$\mathbf{v} = \alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2 + \dots + \alpha_n \mathbf{u}_n, \quad \alpha_1, \dots, \alpha_n \in \mathbb{K}.$$

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Example. Is $(1, 4, 7)$ a linear combination of $(1, 2, 3)$ and $(0, 1, 2)$?

$$\alpha(1, 2, 3) + \beta(0, 1, 2) = (1, 4, 7) \iff \begin{cases} \alpha = 1 \\ 2\alpha + \beta = 4 \\ 3\alpha + 2\beta = 7 \end{cases} \iff \alpha = 1, \beta = 2.$$

Yes: $(1, 4, 7) = 1(1, 2, 3) + 2(0, 1, 2)$.

In general it is a system: if it is inconsistent, the vector is not a linear combination.

Definition and criterion

$V \subseteq U$ is a **subspace** of U if it is a vector space with the *same* operations as U . It suffices to check:

1. $\mathbf{0} \in V$ (in particular, $V \neq \emptyset$);
2. $\mathbf{u}, \mathbf{v} \in V \Rightarrow \mathbf{u} + \mathbf{v} \in V$;
3. $\alpha \in \mathbb{K}, \mathbf{u} \in V \Rightarrow \alpha \mathbf{u} \in V$.

Equivalent criterion (in a single condition): $V \neq \emptyset$ and $\mathbf{u}, \mathbf{v} \in V, \alpha, \beta \in \mathbb{K} \Rightarrow \alpha \mathbf{u} + \beta \mathbf{v} \in V$.

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- $\{(x, y, z) : x + y - z = 0\}$ is a subspace of $\mathbb{R}^3$ (a plane through the origin).
- $\{(x, y, z) : x + y - z = 1\}$ is not: it does not contain $\mathbf{0}$.

Every space U has the subspaces $\{\mathbf{0}\}$ and U .

Definition

The **subspace spanned** by $S = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is the set of all linear combinations:

$$L(S) = \{\alpha_1 \mathbf{u}_1 + \dots + \alpha_n \mathbf{u}_n : \alpha_i \in \mathbb{K}\}.$$

$L(S)$ is always a subspace; S is called a **spanning set** of $L(S)$. Convention: $L(\emptyset) = \{\mathbf{0}\}$.

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Example. $L\{(1, 1, 0), (0, 1, 1)\}$:

$$\alpha(1, 1, 0) + \beta(0, 1, 1) = (\alpha, \alpha + \beta, \beta) = (x, y, z) \iff y = x + z.$$

$L\{(1, 1, 0), (0, 1, 1)\} = \{(x, y, z) : y = x + z\}$, a plane through the origin.

Method: write the generic vector as a linear combination and eliminate the coefficients.

- **Ex. 19–21** linear combinations; in 21 b), describe $L\{\mathbf{u}, \mathbf{v}\}$ by an equation.
- **Ex. 22–23** show that a set is a subspace (criterion).
- **Ex. 24–26** spanned subspaces. In 24 c), note that one of the vectors is the sum of the other two.

2.3 Linear independence

Linear dependence and independence

Definition

$\mathbf{u}_1, \dots, \mathbf{u}_n$ are **linearly independent** (lin. indep.) if

$$\alpha_1 \mathbf{u}_1 + \dots + \alpha_n \mathbf{u}_n = \mathbf{0} \implies \alpha_1 = \dots = \alpha_n = 0.$$

Otherwise (there is a solution with some $\alpha_i \neq 0$), they are **linearly dependent** (lin. dep.).

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How to test: the equation gives a homogeneous system (it always has the zero solution).

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- *infinitely many* solutions $\Rightarrow$ lin. dep.

Example. $(1, 2, 3), (4, 5, 6), (7, 8, 9)$ are lin. dep.: $(7, 8, 9) = 2(4, 5, 6) - (1, 2, 3)$.

Let $\mathbf{u}_1, \dots, \mathbf{u}_n$ be vectors of U .

- I They are lin. dep. $\iff$ one of them is a linear combination of the others.
- II If one of them is $\mathbf{0}$, they are lin. dep.
- III If they are lin. indep., any subset of them is also lin. indep.
- IV Multiplying one of them by $\alpha \neq 0$ does not change independence.
- V Adding a multiple of one of them to another does not change independence.

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- V Adding a multiple of one of them to another does not change independence.

IV and V justify the elimination method: swapping, multiplying by $\alpha \neq 0$ and adding multiples do not change independence.

Example. $\mathbf{u}$, $\mathbf{v}$, $3\mathbf{u} + \mathbf{v}$ are always lin. dep. (Theorem I).

Example: three vectors of $\mathbb{R}^3$

$$\mathbf{c}_1 = (2, -6, -6), \quad \mathbf{c}_2 = (0, 2, 0), \quad \mathbf{c}_3 = (0, -3, -1).$$

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$$\alpha \mathbf{c}_1 + \beta \mathbf{c}_2 + \gamma \mathbf{c}_3 = \mathbf{0}:$$

$$\begin{cases} 2\alpha = 0 \\ -6\alpha + 2\beta - 3\gamma = 0 \\ -6\alpha - \gamma = 0 \end{cases} \Rightarrow \alpha = 0, \gamma = 0, \beta = 0.$$

$\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ are lin. indep. We will come back to these three vectors in the next chapters.

- **Ex. 27–29** test independence (homogeneous system).
- **Ex. 30** the same, *without calculations*: use Theorems I–III.
- **Ex. 31** abstract lin. indep. vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$: use the definition or the theorems (there are no coordinates).

2.4 Basis and dimension

Definition

A **basis** of U is a linearly independent spanning set of U .

Theorem

If U has a finite basis, all bases of U have the same number of vectors: the **dimension** of U , $\dim U$. ($\dim\{\mathbf{0}\} = 0$: the basis is $\emptyset$.)

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Space	Standard basis	Dimension
$\mathbb{R}^n$	$\mathbf{e}_1 = (1, 0, \dots, 0), \dots, \mathbf{e}_n = (0, \dots, 0, 1)$	n
P_n	$1, x, x^2, \dots, x^n$	$n + 1$
$\mathbb{C}$ over $\mathbb{R}$	$1, i$	2

Theorem

Let $\dim U = n$.

1. Any $n + 1$ (or more) vectors of U are lin. dep.
2. Any n lin. indep. vectors form a basis.
3. Any n vectors that span U form a basis.

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Example. $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ (Section 2.3) are 3 lin. indep. vectors in $\mathbb{R}^3 \Rightarrow$ basis of $\mathbb{R}^3$.

Example. $A = \{(x, y, z, w) \in \mathbb{R}^4 : x + y + z + w = 0\}.$

Subspaces given by equations

Example. $A = \{(x, y, z, w) \in \mathbb{R}^4 : x + y + z + w = 0\}.$

$$x = -y - z - w \Rightarrow (x, y, z, w) = y(-1, 1, 0, 0) + z(-1, 0, 1, 0) + w(-1, 0, 0, 1).$$

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Rule of thumb

In $\mathbb{R}^n$, a subspace given by k **independent** homogeneous linear equations has dimension $n - k$: one free coordinate for each basis vector.

- **Ex. 32** basis of $\mathbb{R}^2$.
- **Ex. 33** a spanning set that is not a basis: remove the extra vector.
- **Ex. 34** basis of P_2 , in two ways (with and without the dimension).
- **Ex. 35–37** basis and dimension of subspaces. In 35 b), careful: three spanning vectors do not guarantee dimension 3.

2.5 Coordinates and change of basis

Theorem and definition

If $B = (\mathbf{u}_1, \dots, \mathbf{u}_n)$ is an *ordered* basis of U , each $\mathbf{v} \in U$ can be written in a **unique** way as

$$\mathbf{v} = \lambda_1 \mathbf{u}_1 + \dots + \lambda_n \mathbf{u}_n.$$

$(\lambda_1, \dots, \lambda_n)$ are the **coordinates** of $\mathbf{v}$ in the basis B : $[\mathbf{v}]_B = (\lambda_1, \dots, \lambda_n)$.

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Example. $B = ((1, 1), (1, 2))$, $\mathbf{v} = (3, 5)$:

$$\lambda_1(1, 1) + \lambda_2(1, 2) = (3, 5) \iff \lambda_1 = 1, \lambda_2 = 2. \quad [\mathbf{v}]_B = (1, 2).$$

In the standard basis, the coordinates are the components themselves: $(3, 5)$.

The same vector has different coordinates in different bases:

- standard basis: $(3, 5)$;
- $B = ((1, 1), (1, 2))$: $(1, 2)$;
- $B' = ((1, 2), (1, 1))$ (same vectors, different order): $(2, 1)$.

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Method. To go from one basis to another:

1. rebuild the vector from the known coordinates;
2. solve the system that gives the coordinates in the new basis.

In Ch. 3 we will see a systematic way of doing these calculations.

- **Ex. 38** coordinates in bases of P_2 . In c), the bases D and E have the same vectors in a different order: compare the coordinates.

Measurements: $(0, 1)$, $(1, 3)$, $(2, 7)$. We look for $p \in P_2$ with $p(0) = 1$, $p(1) = 3$, $p(2) = 7$.

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In the standard basis $(1, x, x^2)$: $p = a + bx + cx^2$ gives a system,

$$a = 1, \quad a + b + c = 3, \quad a + 2b + 4c = 7 \Rightarrow p = 1 + x + x^2.$$

LINEAR ALGEBRA IN PRACTICE: interpolation

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In the Lagrange basis (L_0, L_1, L_2) , with $L_i(x_j) = 1$ if $i = j$ and 0 if $i \neq j$:

$$L_0 = \frac{(x-1)(x-2)}{2}, \quad L_1 = -x(x-2), \quad L_2 = \frac{x(x-1)}{2}, \quad p = 1 L_0 + 3 L_1 + 7 L_2.$$

The coordinates in the Lagrange basis are the measurements themselves — no system needs to be solved.

Choosing the basis well can make the calculations trivial. This idea recurs several times in the course.

Concept	How to obtain it / what it says
Vector space	addition and scalar multiplication satisfying the 8 axioms
Not a vector space	one failing axiom, with an example, is enough
Linear combination	$\alpha_1 \mathbf{u}_1 + \cdots + \alpha_n \mathbf{u}_n$; deciding means solving a system
Subspace	contains $\mathbf{0}$, closed under $+$ and under scalar multiplication
$L(S)$	all linear combinations of S ; described by equations
lin. indep.	$\sum \alpha_i \mathbf{u}_i = \mathbf{0}$ only with all $\alpha_i = 0$ (unique solution)
lin. dep.	some vector is a linear combination of the others; or there is a $\mathbf{0}$
Basis	lin. indep. spanning set; dimension = no. of vectors in a basis
In dimension n	n lin. indep. $\Rightarrow$ basis; n spanning $\Rightarrow$ basis; $n + 1$ vectors $\Rightarrow$ lin. dep.
Subspace by equations	dimension = $n - (\text{no. of independent equations})$
Coordinates	unique in an ordered basis; change with the basis and with the order

KEY TAKEAWAYS — Chapter 2

1. A vector space is any set where we add and multiply by scalars with the usual rules: n -tuples, polynomials and functions are examples of the same concept.
2. A subspace is a subset that is closed under these two operations; hence it always contains the zero vector.
3. Linearly independent vectors have no redundancy: none of them can be obtained from the others.
4. A basis is a spanning set without redundancy; all bases have the same number of vectors, the dimension.
5. In a basis, each vector has unique coordinates; changing the basis changes the coordinates, not the vector.

Next (Ch. 3): matrices — the tool that makes these calculations systematic.

References (full list: course bibliography)

- Santana, A. P., Queiró, J. F., *Introdução à Álgebra Linear*, Gradiva (in Portuguese).
- Lipschutz, S., Lipson, M., *Schaum's Outline of Linear Algebra*, McGraw-Hill — many solved exercises.
- Strang, G., *Introduction to Linear Algebra*, Wellesley-Cambridge Press.

Resources — Sheet 2 (exercises 16–38) and its solutions.